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Hiring Committee
xAI
Stanford Research Park
Palo Alto, CA 94304

Dear Hiring Committee,

I am writing to express my interest in the Research Scientist position at xAI. I am a Ph.D. candidate in the Department of Economics at Johns Hopkins University (advisors: Christopher D. Carroll, Nicholas W. Papageorge, and Stelios Fourakis), and I expect to receive my degree in May 2026. My primary specializations are macroeconomic modeling, wealth and income dynamics, and computational economics.

My research agenda focuses on the ability of macroeconomic models to generate empirically realistic distributions of wealth and income over the life cycle. A key driver of that work is modeling heterogeneity in the rate of return on assets, and understanding how return dispersion contributes to portfolio accumulation across households. These themes align well with xAI's focus on building systems that reason clearly, model complex dynamics, and learn from data in transparent, interpretable, and scientifically grounded ways.

In my job-market paper, I calibrate a life-cycle heterogeneous-agent model to match wealth moments from the Survey of Consumer Finances by allowing households to differ in their portfolio returns. The model helps explain observed wealth concentration and return dispersion. In a related research project using the Health and Retirement Study, I examine how trust influences both income growth and asset returns—finding a hump-shaped relationship for returns and characterizing the distribution of the persistent component of returns to net worth. These experiences have equipped me to: (i) frame research strategies and design modeling approaches; (ii) execute econometric and simulation analysis with large micro-data sets; and (iii) translate results into clear, rigorous written deliverables for both academic and practitioner audiences.

I am particularly drawn to xAI's mission of building advanced AI systems grounded in mathematical rigor and deep reasoning. My background in simulation-based modeling, heterogeneous-agent dynamics, and empirical validation makes me excited about contributing to problems involving large-scale optimization, data distribution drift, agent-based modeling, and generalization. I am especially motivated by the opportunity to collaborate with world-class researchers and engineers to develop systems that better understand complex human behavior, incentives, and decision-making environments.

I believe I would be a strong fit at xAI because:

1. My experience with structural modeling, simulation, and statistical inference aligns with xAI's work on developing models that learn complex patterns and reason from first principles.

2. My research on heterogeneity and dynamic behavior prepares me to contribute to modeling real-world agents, distributions, incentives, and environments relevant to scalable AI research.
3. I bring strong communication skills and a collaborative mindset—essential for contributing to interdisciplinary research teams working at the frontier of AI and quantitative modeling.

I would welcome the opportunity to contribute my analytical rigor, modeling experience, and research insights to xAI's mission. Thank you for considering my application.

Sincerely,

Decory Edwards